FROBENIUS ELEMENTS AT FINITELY MANY CLOSED POINTS TOPOLOGICALLY GENERATE THE FUNDAMENTAL GROUP

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Riemann hypothesis for X and for Y: #X(k_N) lies between q^N + 1 - 2g_X q^{N/2} and q^N + 1 + 2g_X q^{N/2}, and similarly for Y. invoked at proof lines 4186-4192. Provides the point-count bounds that force G trivial.
- Riemann-Hurwitz for the finite etale cover: 2g_Y - 2 = #G (2g_X - 2). invoked at proof line 4193. Enters the genus in the final inequality.
- R.H. for X gives points of every degree up to N among the chosen set, making the degree map on Gamma surjective. invoked at proof lines 4173-4176. Shows Y_kbar is irreducible, i.e. U maps onto the full degree group.

Proposition 0.2. *There exists a finite set $x_1, \ldots, x_n$ of closed points of X such that set of* **all** *frobenius elements corresponding to these points topologically generate $\pi_1(X)$.*

Proof. Pick $N \gg 0$ and let

$$\{x_1, \ldots, x_n\} = \begin{array}{l}\text{set of all closed points of} \\ X \text{ of degree} \leq N \text{ over } k\end{array}$$

Let $\Gamma \subset \pi_1(X)$ be as in the variant statement for these points. Assume $\Gamma \neq \pi_1(X)$. Then we can pick a normal open subgroup U of $\pi_1(X)$ containing Γ with $U \neq \pi_1(X)$. By R.H. for X our set of points will have some x_{i_1} of degree N, some x_{i_2} of degree $N-1$. This shows $\deg : \Gamma \to \widehat{\mathbf{Z}}$ is surjective and so the same holds for U. This exactly means if $Y \to X$ is the finite étale Galois covering corresponding to U, then $Y_{\overline{k}}$ irreducible. Set $G = \mathrm{Aut}(Y/X)$. Picture

$$Y \to^G X, \quad G = \pi_1(X)/U$$

By construction all points of X of degree $\leq N$, split completely in Y. So, in particular

$$\#Y(k_N) \geq (\#G)\#X(k_N)$$

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Use R.H. on both sides. So you get

$$q^N + 1 + 2g_Y q^{N/2} \geq \#G \#X(k_N) \geq \#G(q^N + 1 - 2g_X q^{N/2})$$

Since $2g_Y - 2 = (\#G)(2g_X - 2)$, this means

$$q^N + 1 + (\#G)(2g_X - 1) + 1)q^{N/2} \geq \#G(q^N + 1 - 2g_X q^{N/2})$$

Thus we see that G has to be the trivial group if N is large enough. $\square$